Informational Commands

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You can enter "--help" after any of the commands listed below to show their help page in your terminal window

Information on Jobs

The **squeue** command provides information about jobs running on the analysis cluster in the following format: Job ID, Partition, Name, User, Job State, Time, Nodes, Reason

```
squeue                          //lists all current (and pending) jobs in the queue
squeue -u <username>            //lists all jobs in the queue for the specified user
squeue -u <username> -t PENDING //lists all pending jobs for the specified user
squeue -u <username> -t RUNNING //lists all running jobs for the specified user
```

It is important to note that the order in which jobs are printed on the screen are only the order of submission. Execution occurs based on a job priority algorithm called fairshare.

Sample output:

```
[user@submit0 ~]$ squeue
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            191407  c2_short jobs-con rkuplick PD       0:00      1 (BeginTime)
            190674    c2_cpu jobs-dai rkuplick PD       0:00      1 (BeginTime)
            189833    c2_cpu jobs_dai jtouthan PD       0:00      1 (BeginTime)
            191355    c2_cpu    t1kfc  nkirlic  R      35:49      1 compute205
            191404    c2_cpu BB488-T0 bdoudica  R       3:20      1 compute202
            191382    c2_cpu ML-HB_sl   kclary  R      23:42      1 compute206
[user@submit0 ~]$ squeue -u rkuplicki
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            191408  c2_short jobs-con rkuplick PD       0:00      1 (BeginTime)
            190674    c2_cpu jobs-dai rkuplick PD       0:00      1 (BeginTime)
[user@submit0 ~]$ squeue -u rkuplicki -t PENDING
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            191408  c2_short jobs-con rkuplick PD       0:00      1 (BeginTime)
            190674    c2_cpu jobs-dai rkuplick PD       0:00      1 (BeginTime)
[user@submit0 ~]$ squeue -u rkuplicki -t RUNNING
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
             // Nothing displayed here because no jobs were running for this user
```

For additional information refer to the official documentation at: Slurm Workload Manager - squeue

Partition Information

The **sinfo** command provides information about the partitions (or queues) you have access to use.

```
sinfo                       //provides information on all the queues you are assigned a
sinfo -p <partition name>   //provides information for the specified queue
```

The information is displayed in the following format: Partition Name, Availability, Time Limit, Nodes, State, Node List.

The most common states you will see are defined as:

- Allocated - The node has been allocated to one or more jobs.
- Down - The node is unavailable for use.
- Drain - The node is finishing the jobs it has allocated. Once finished, the node will not be allocated to other jobs until taken out of drain.
- Fail - The node is failing or expected to fail soon and is unavailable.
- Idle - The node is not allocated to any jobs and is ready for use.
- Mixed - The node has some of its CPUs allocated while others are idle
- Reserved - The node is in an advanced reservation and not generally available.

Sample output:

```
[user@submit0 ~]$ sinfo
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
c2_cpu       up 3-00:00:00      1  drain compute207
c2_cpu       up 3-00:00:00      5    mix compute[200-203,206]
c2_cpu       up 3-00:00:00      1  alloc compute205
c2_cpu       up 3-00:00:00      1   idle compute204
c2_short*    up    2:00:00      1  drain compute207
c2_short*    up    2:00:00      5    mix compute[200-203,206]
c2_short*    up    2:00:00      1  alloc compute205
c2_short*    up    2:00:00      1   idle compute204
c2_gpu       up 3-00:00:00      1  drain compute207
c2_gpu       up 3-00:00:00      1    mix compute206
c2_gpu       up 3-00:00:00      1  alloc compute205
c2_gpu       up 3-00:00:00      1   idle compute204
[user@submit0 ~]$ sinfo -p c2_cpu
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
c2_cpu       up 3-00:00:00      1  drain compute207
c2_cpu       up 3-00:00:00      5    mix compute[200-203,206]
c2_cpu       up 3-00:00:00      1  alloc compute205
c2_cpu       up 3-00:00:00      1   idle compute204
```

For additional information refer to the official documentation at: https://slurm.schedmd.com/sinfo.html

Debugging Jobs

The **scontrol** command can be used for getting configuration details about a job, node, or partition. It is especially useful when debugging jobs.

```
scontrol show job <job number>  //gives details about the particular job
scontrol show partition         //provides configuration details (example: priority, al
```

Sample output:

```
[user@submit0 ~]$ scontrol show job 191832
JobId=191832 JobName=BB710-T1-preprocess-functional_session-_MID.bash
   UserId=bmathis(1715003793) GroupId=domain users(1715000513) MCS_label=N/A
   Priority=5141 Nice=0 Account=(null) QOS=normal
   JobState=RUNNING Reason=None Dependency=(null)
   Requeue=1 Restarts=0 BatchFlag=1 Reboot=0 ExitCode=0:0
   RunTime=00:17:51 TimeLimit=1-00:00:00 TimeMin=N/A
   SubmitTime=2019-06-25T10:20:37 EligibleTime=2019-06-25T10:20:37
   AccrueTime=Unknown
   StartTime=2019-06-25T10:20:38 EndTime=2019-06-26T10:20:38 Deadline=N/A
   PreemptTime=None SuspendTime=None SecsPreSuspend=0
   LastSchedEval=2019-06-25T10:20:38
   Partition=c2_cpu AllocNode:Sid=submit0:2470532
   ReqNodeList=(null) ExcNodeList=(null)
   NodeList=compute204
   BatchHost=compute204
   NumNodes=1 NumCPUs=15 NumTasks=15 CPUs/Task=1 ReqB:S:C:T=0:0:*:*
   TRES=cpu=15,mem=60000M,node=1,billing=15
   Socks/Node=* NtasksPerN:B:S:C=0:0:*:* CoreSpec=*
   MinCPUsNode=10 MinMemoryNode=60000M MinTmpDiskNode=0
   Features=(null) DelayBoot=00:00:00
   OverSubscribe=OK Contiguous=0 Licenses=(null) Network=(null)
   Command=(null)
   WorkDir=/media/cephfs/labs/raupperle/TreatmentStudy/Data/Data_Analysis/processing-f
   StdErr=/media/cephfs/labs/raupperle/TreatmentStudy/Data/Data_Analysis/processing-fi
   StdIn=/dev/null
   StdOut=/media/cephfs/labs/raupperle/TreatmentStudy/Data/Data_Analysis/processing-fi
   Power=
[user@submit0 ~]$ scontrol show partition
PartitionName=c2_cpu
   AllowGroups=ALL AllowAccounts=ALL AllowQos=ALL
   AllocNodes=ALL Default=NO QoS=N/A
   DefaultTime=1-00:00:00 DisableRootJobs=NO ExclusiveUser=NO GraceTime=0 Hidden=NO
   MaxNodes=UNLIMITED MaxTime=3-00:00:00 MinNodes=0 LLN=NO MaxCPUsPerNode=20
   Nodes=compute[200-207]
   PriorityJobFactor=1 PriorityTier=10 RootOnly=NO ReqResv=NO OverSubscribe=FORCE:1
   OverTimeLimit=NONE PreemptMode=SUSPEND
   State=UP TotalCPUs=160 TotalNodes=8 SelectTypeParameters=NONE
   JobDefaults=(null)
   DefMemPerCPU=6000 MaxMemPerCPU=6000
PartitionName=c2_short
   AllowGroups=ALL AllowAccounts=ALL AllowQos=ALL
   AllocNodes=ALL Default=YES QoS=c2_short_qos
   DefaultTime=02:00:00 DisableRootJobs=NO ExclusiveUser=NO GraceTime=0 Hidden=NO
   MaxNodes=UNLIMITED MaxTime=02:00:00 MinNodes=0 LLN=NO MaxCPUsPerNode=20
   Nodes=compute[200-207]
   PriorityJobFactor=2 PriorityTier=20 RootOnly=NO ReqResv=NO OverSubscribe=NO
   OverTimeLimit=NONE PreemptMode=OFF
   State=UP TotalCPUs=160 TotalNodes=8 SelectTypeParameters=NONE
   JobDefaults=(null)
   DefMemPerCPU=6000 MaxMemPerCPU=6000
PartitionName=c2_gpu
   AllowGroups=ALL AllowAccounts=ALL AllowQos=ALL
   AllocNodes=ALL Default=NO QoS=N/A
   DefaultTime=1-00:00:00 DisableRootJobs=NO ExclusiveUser=NO GraceTime=0 Hidden=NO
   MaxNodes=UNLIMITED MaxTime=3-00:00:00 MinNodes=0 LLN=NO MaxCPUsPerNode=20
   Nodes=compute[204-207]
   PriorityJobFactor=3 PriorityTier=30 RootOnly=NO ReqResv=NO OverSubscribe=NO
   OverTimeLimit=NONE PreemptMode=OFF
   State=UP TotalCPUs=80 TotalNodes=4 SelectTypeParameters=NONE
   JobDefaults=(null)
   DefMemPerCPU=6000 MaxMemPerCPU=6000
```

For additional information refer to the official documentation at: https://slurm.schedmd.com/scontrol.html